APPENNINO RPG ASSET FACTORY VISUAL REVIEW DOSSIER

Comprehensive Technical Evidence Portfolio for Multimodal AI Review & Peer Audit

INSPECTION METHODOLOGY NOTE FOR MULTIMODAL AI AGENTS (e.g. ChatGPT):

This dossier contains uncompressed, un-retouched rendering evidence from both experiments: 3D Procedural Blender and 2D AI-Native.

All dimensions, scale anchors (96 PPM, 202px/2.1m doors), and perspective lines are factual and derived programmatically.

Repository: https://github.com/AhmadTahmid/apennino-rpg-asset-factory (Branch: main)

GROUND TRUTH REFERENCE

Reference Target Frame

Pipeline: Extracted Video Frame (1 fps) | Original Resolution: 1280x720

Source: C:\Users\Acer\Downloads\A_random_Appennino_Italian_tow.mp4 (frame_001.jpg)

Aesthetic Target: Warm Italian Appennino hill-town architecture, weathered limestone, terracotta barrel roofs, olive shutters, vertical stairs.

Perspective: Dimetric 3/4 elevated angle (~26.5 deg ground slope). Sunlight from top-left. Model: None (Ground Truth).

OVERVIEW

Master Comparative Contact Sheet (8 Pipelines)

Pipeline: Multi-Pipeline Overview | Original Resolution: 2680x1192

APPENNINO RPG ASSET FACTORY EXPERIMENT COMPARISON CONTACT SHEET

Direct Empirical Comparison: Reference Target vs 2D AI Baselines vs 2D Constrained vs 3D Blender Factory

01. REFERENCE TARGET

Original Appennino hill-town video frame

02. RAW AI BASELINE

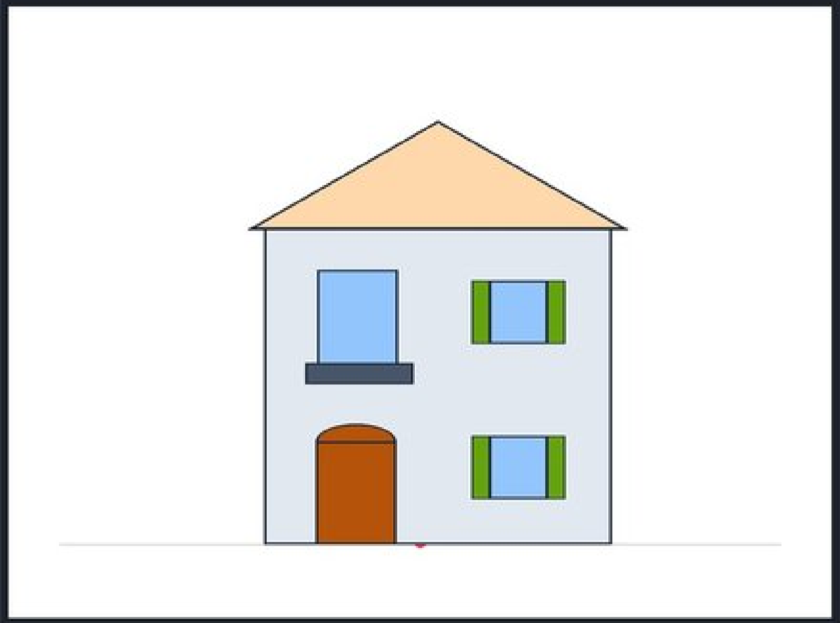
Prompt-only generation (Gemini 3.1 Flash Image)

03. STYLE-ANCHORED AI

Conditioned on Style Anchor Board

04. STRUCTURAL SCAFFOLD

Deterministic 2:1 dimetric vector blueprint



05. NATIVE 2D COMPOSITE

Metric anchor composition (Zero eyeballed resize)

06. HYBRID SCENE

2D structured assets + generative painterly backdrop

07. BLENDER 3D CLEAN

Procedural Cycles 3D->2D render (100% locked camera)

08. BLENDER 3D PAINTERLY

Procedural 3D render with line contour post-pass

Side-by-side comparison across all major pipelines: Reference Video Frame, Raw AI Baseline, Style-Anchored AI, Structural Blueprint,

Native 2D Composite, Hybrid Generative Scene, and Procedural Blender 3D Renders (Clean & Painterly).

Repository Path: review_bundle/contact_sheet.jpg

MODULAR ASSETS

Modular Transparent Asset Grid (8 Sprites)

Pipeline: Automated Normalization | Original Resolution: 2148x1254

MODULAR ASSETS CONTACT SHEET □ INDIVIDUAL TRANSPARENT SPRITES

Extracted with BFS flood-fill alpha, normalized to canonical 96 PPM & Y=900 baseline anchor

HOUSE A (2-FLOOR)

Gable roof, olive shutters, balcony, ivy

HOUSE B (3-FLOOR TOWER)

Narrow tower, brown shutters, chimney

HOUSE C (WIDE SHOP)

Hip roof, arched shop door, flower boxes

STONE STAIRS

Steps climbing 2.4m elevation with parapet

RETAINING WALL

Limestone terrace wall with coping stones

CARVED FOUNTAIN

Octagonal basin with center column finial

MEDITERRANEAN VEGETATION

Olive tree, garden shrubs, potted geraniums

STREET PROPS

Barrels, stacked fruit crates, wall lantern

Individual sprites extracted via BFS flood-fill alpha defringing. Preserves soft contact shadows and removes solid white backgrounds.

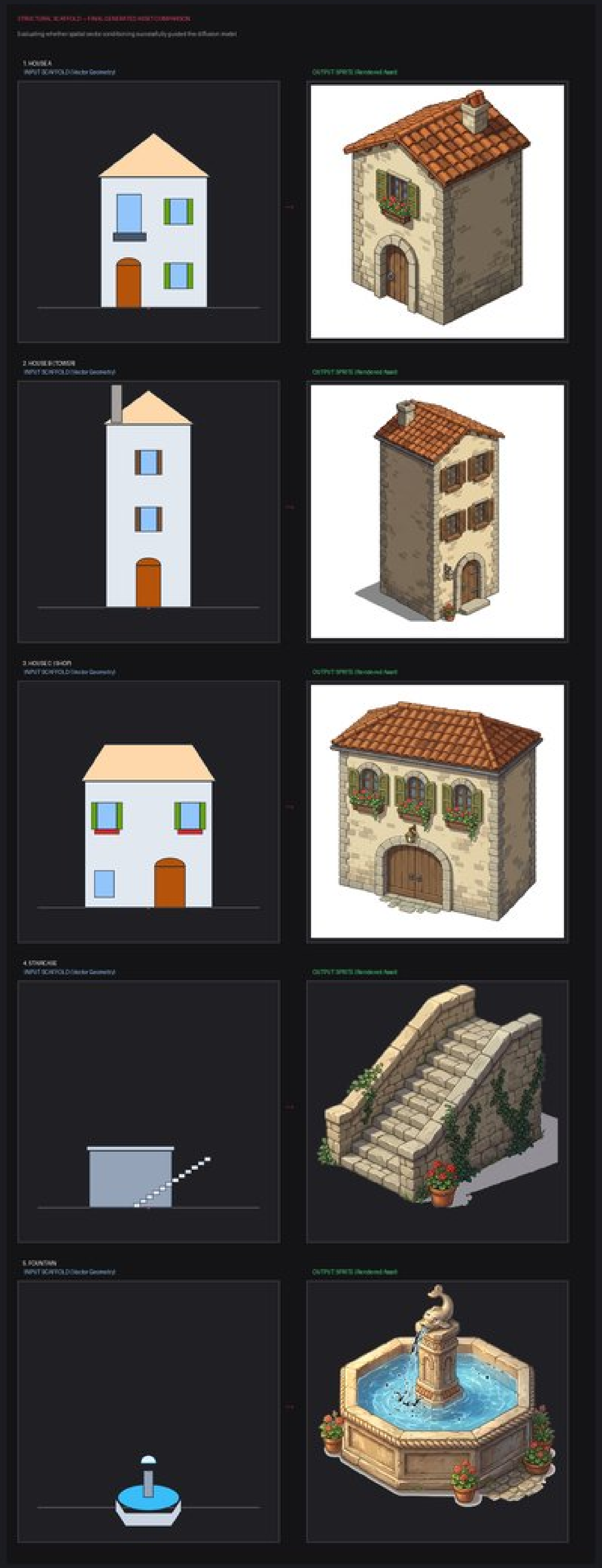
Canonical Scale: 96 PPM. Ground contact anchor aligned to (X=512, Y=900) on 1024x1024 canvas.

Repository Path: review_bundle/assets_contact_sheet.png

SPATIAL CONDITIONING

Scaffolds vs. Outputs Comparison Sheet

Pipeline: Deterministic Blueprint vs Output | Original Resolution: 1060x2770



Evaluates whether spatial vector blueprints (blue outlines, 2:1 dimetric grid) effectively guided the diffusion model.

Pairs shown: House A, House B (Tower), House C (Shop), Stairs, and Fountain.

Repository Path: review_bundle/scaffolds_contact_sheet.png

CONTROL GROUP A

Raw AI Baseline Generation (House A)

Pipeline: gemini-3.1-flash-image (Prompt-Only) | Original Resolution: 1024x1024

Prompt-only + video reference conditioning without structural geometric scaffold.

Observations: Rich surface rendering, attractive terracotta roof corrugation, but scale is uncontrolled (macro framing drift).

Repository Path: review_bundle/02_raw_baseline.png

STYLE CONDITIONING

Unified Visual Style Anchor Board

Pipeline: Multimodal Reference Anchor | Original Resolution: 1200x900

	Limestone Base
	Stone Shadow
	Terracotta Tile
	Roof Highlight
	Olive Shutter
	Aged Wood
	Fountain Water
	Geranium Red
	Climbing Ivy
	Mountain Haze

APPENNINO RPG 2D ASSET FACTORY - STYLE ANCHOR BOARD

Dimetric 3/4 Perspective | Warm Limestone | Terracotta Barrel Roofs | Olive Shutters

Combines representative crops of limestone masonry, stairs, fountain, and calibrated hex color palette swatches.

Enforces color consistency (olive green #4E6B4A, terracotta #C75A2A, limestone #D8D1C4) across separate generations.

Repository Path: appennino-2d-asset-factory/style/anchor_board.png

CONTROL GROUP B

Style-Anchored Generation (House A)

Pipeline: gemini-3.1-flash-image + Anchor Board | Original Resolution: 1024x1024

Conditioned explicitly on style/anchor_board.png.

Observations: Noticeable improvement in color harmony; exact terracotta hue, exposed rafter tails, and olive shutter tones match.

Repository Path: review_bundle/03_style_anchored.png

NATIVE 2D ASSEMBLY

Native 2D Town Composite (Zero Eyeballed Scaling)

Pipeline: Programmatic 2D Depth Layering | Original Resolution: 1920x1080

Assembled from independently generated 2D sprites positioned at canonical world anchor coordinates.

Rules: No arbitrary eyeballed resizing. Scaled strictly via 202px (2.1m) doorway anchor. No rotation or warping applied.

Repository Path: review_bundle/06_native_composite.png

DIAGNOSTIC OVERLAY

Native Composite Semantic Telemetry Overlay

Pipeline: Analytical Coordinate Overlay | Original Resolution: 1920x1080

Red Pins: Canonical ground baseline contact anchors (X, Y). Green Bars: Verified 2.1m doorway heights (202px at 96 PPM).

Dashed Line: Elevation terrace boundary. Numbers [1]..[8]: Back-to-front depth sorting sequence.

Repository Path: review_bundle/native_composite_diagnostic.png

HYBRID COMPOSITION

Hybrid Scene: Structured Town Sprites + Generative Backdrop

Pipeline: Composite + Matte Backdrop | Original Resolution: 1920x1080

Combines native 2D gameplay assets with a generative painterly distant mountain vista.

Key Takeaway: 2D AI models excel dramatically at atmospheric environmental backdrops where geometric collision is irrelevant.

Repository Path: review_bundle/07_hybrid_scene.png

Procedural Blender 3D Clean Cycles Render

Pipeline: Blender 5.2.1 LTS bpy Cycles (64 samples) | Original Resolution: 1920x1080

Built entirely from procedural Python code (architecture.py, materials.py). 100% locked dimetric camera (56 deg pitch, 45 deg yaw).

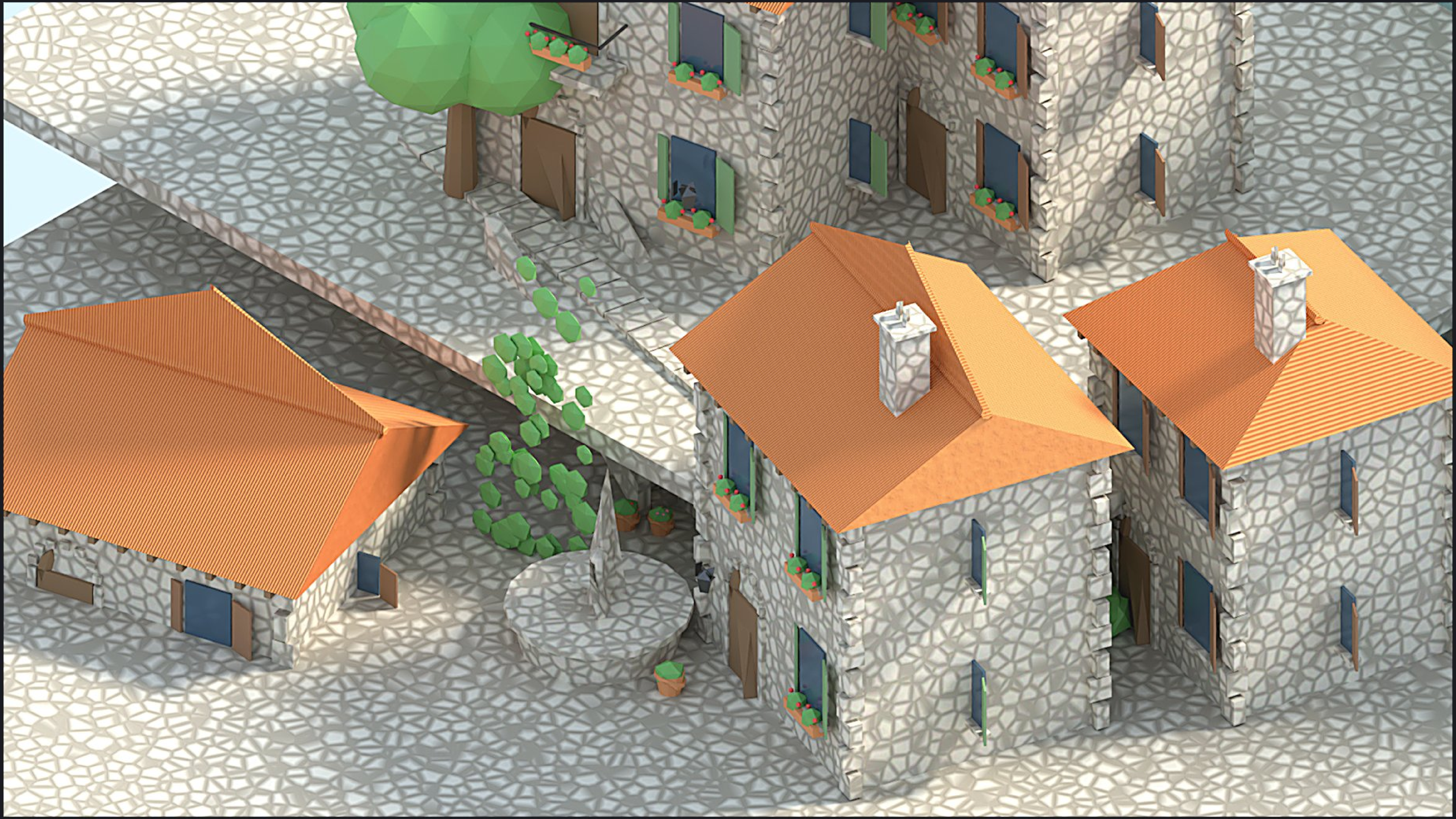
Observations: Zero perspective drift and perfect metric scale truth, but pure mathematical procedural stone looks somewhat sterile.

Repository Path: review_bundle/08_blender_clean.png

BLENDER 3D PIPELINE

Procedural Blender 3D Painterly / Illustrated Variant

Pipeline: Blender Cycles + Ink Contour Filter | Original Resolution: 1920x1080



Procedural 3D render processed through automated line contour filter and warm color grade (scripts/postprocess.py).

Observations: Bridges the gap between 3D CGI and hand-crafted 2D JRPG concept art while preserving 100% spatial precision.

Repository Path: review_bundle/09_blender_painterly.png

Procedural Blender 3D Pixel-ish RPG Variant

Pipeline: 480x270 -> 4x Nearest Neighbor Upscale | Original Resolution: 1920x1080

Simulates retro 16-bit / 32-bit JRPG sprite rendering via integer resolution downscale and nearest-neighbor upscaling.

Repository Path: review_bundle/10_blender_pixelish.png

ASSET AUDIT

Modular Asset: House A (2-Floor Stone House)

Pipeline: Raw Generation vs. Normalized Transparent Sprite | 1024x1024

1. Raw Output (Solid White Background)

2. Normalized Transparent Sprite (BFS Alpha Cleaned)

Left: Raw prompt generation from gemini-3.1-flash-image with solid white background.

Right: Automated background removal, alpha defringing, doorway anchor detection (210px -> 202px, scale factor x0.962).

Canonical Ground Anchor: (X=512, Y=900) on 1024x1024 canvas. World Dimensions: 8.47m width x 9.04m height.

ASSET AUDIT

Modular Asset: House B (3-Floor Narrow Tower)

Pipeline: Raw Generation vs. Normalized Transparent Sprite | 1024x1024

1. Raw Output (Solid White Background)

2. Normalized Transparent Sprite (BFS Alpha Cleaned)

Left: Raw prompt generation showing narrow 3-floor townhouse tower with stone chimney and dark brown shutters.

Right: Automated BFS alpha extraction and metric scaling (detected door 205px -> 202px, scale factor x0.985).

World Dimensions: 10.31m width x 9.56m height. Zero manual repainting.

ASSET AUDIT

Modular Asset House C (2-Floor Wide Townhouse)

Pipeline: Raw Generation vs. Normalized Transparent Sprite | 1024x1024

1. Raw Output (Solid White Background)

2. Normalized Transparent Sprite (BFS Alpha Cleaned)

Left: Wide frontage shop building with arched wooden double doors and flower boxes with blooming red geraniums.

Right: Normalized sprite on transparent canvas (detected door 215px -> 202px, scale factor x0.940).

World Dimensions: 9.64m width x 8.33m height

ASSET AUDIT

Modular Infrastructure: Stairs & Retaining Wall

Pipeline: Transparent Sprites on Checkerboard | 1024x1024

Stone Stairs (Climbing 2.4m Elevation)

Limestone Terrace Retaining Wall

Left: Weathered stone steps climbing from bottom-left to top-right with side stone coping and potted flower.

Right: Massive weathered limestone retaining wall with coping stones and climbing green ivy.

Scale: 1.000 (1024 native scale).

ASSET AUDIT

Modular Props: Piazza Fountain & Street Furniture

Pipeline: Transparent Sprites on Checkerboard | 1024x1024

Octagonal Piazza Carved Stone Fountain

Street Props: Barrels, Crates & Lantern

Left: Carved octagonal stone basin with sparkling blue water and center fountain finial.

Right: Oak barrels with iron hoops, wooden vegetable crates with apples and potatoes, and wrought-iron wall lantern.

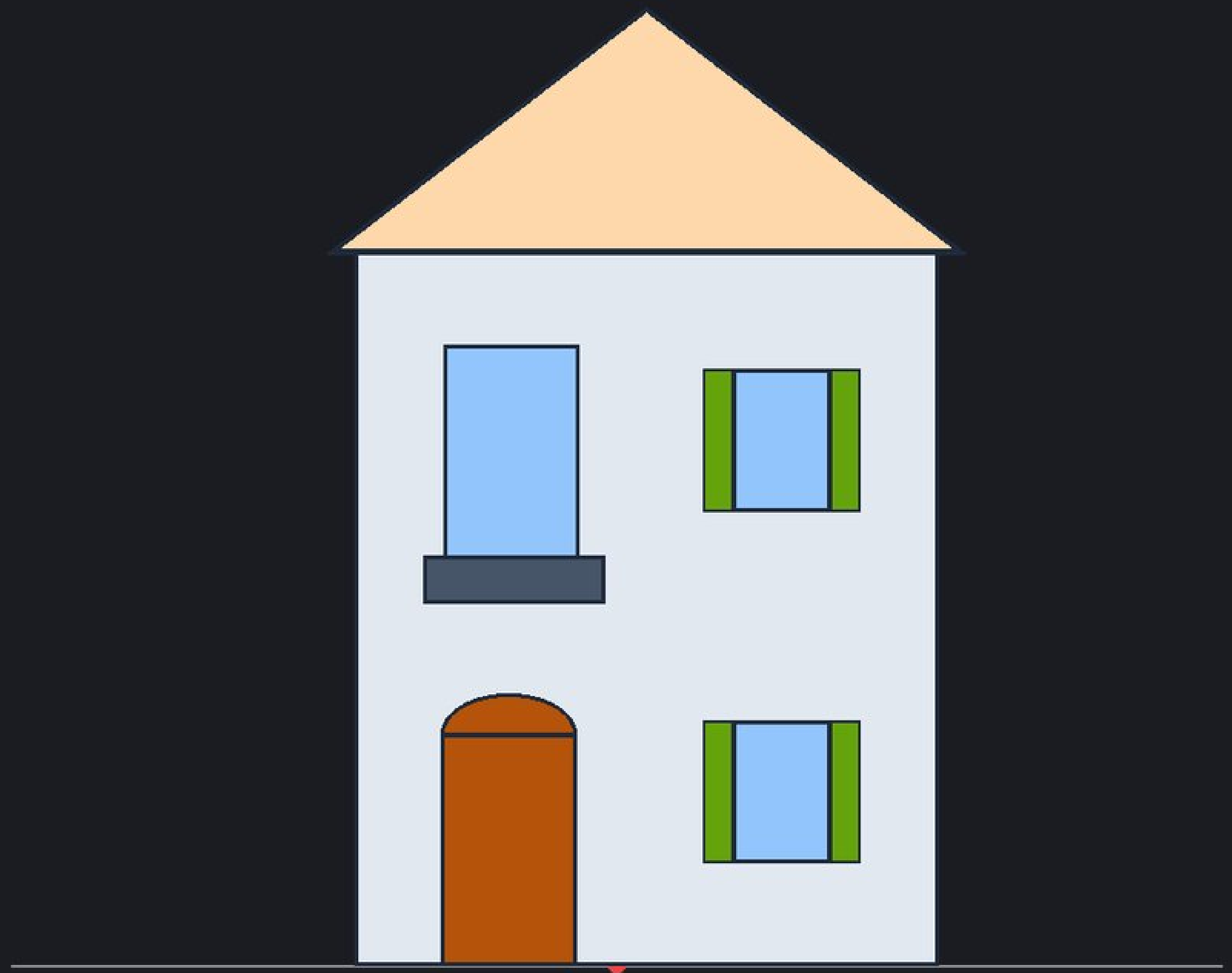
Notice: Props macro framing makes individual crates slightly oversized relative to building doorways without sub-cropping.

SPATIAL CONDITIONING

Scaffold Adherence Test: House A

Pipeline: Vector Blueprint vs. Diffusion Output | 1024x1024

Deterministic Vector Scaffold (Blueprint)



Generated Result (gemini-3.1-flash-image)

Scaffold Dimensions: Width 4.5m (432px), Wall Height 5.6m (538px), Roof Pitch 1.8m (173px), Door 2.1m (202px).

Adherence Assessment: Model respected the 2-storey silhouette, gable roof pitch, and arched door location accurately.

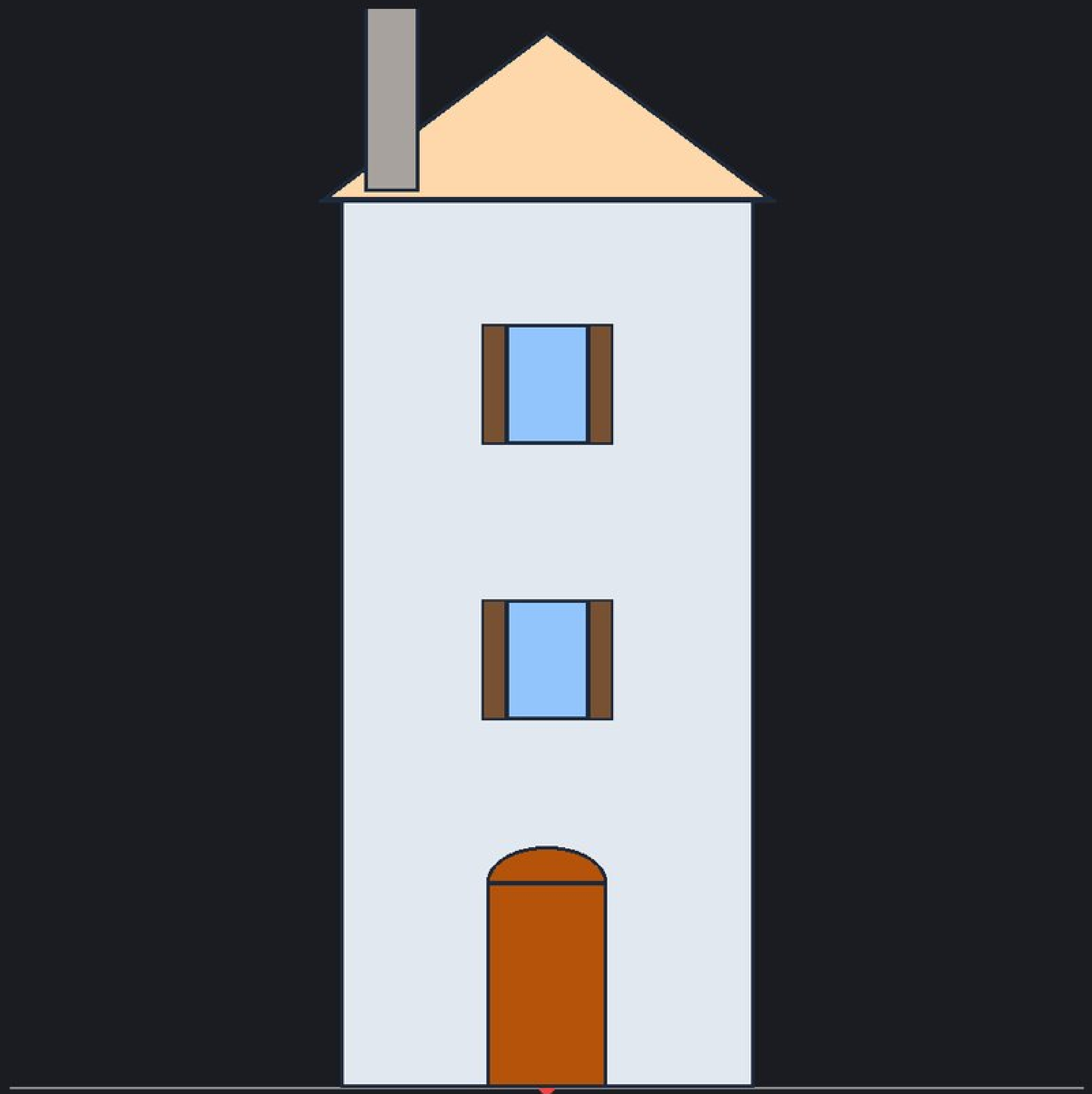
Normalization Factor: x0.962 (Door matched within 4% of canonical specification).

SPATIAL CONDITIONING

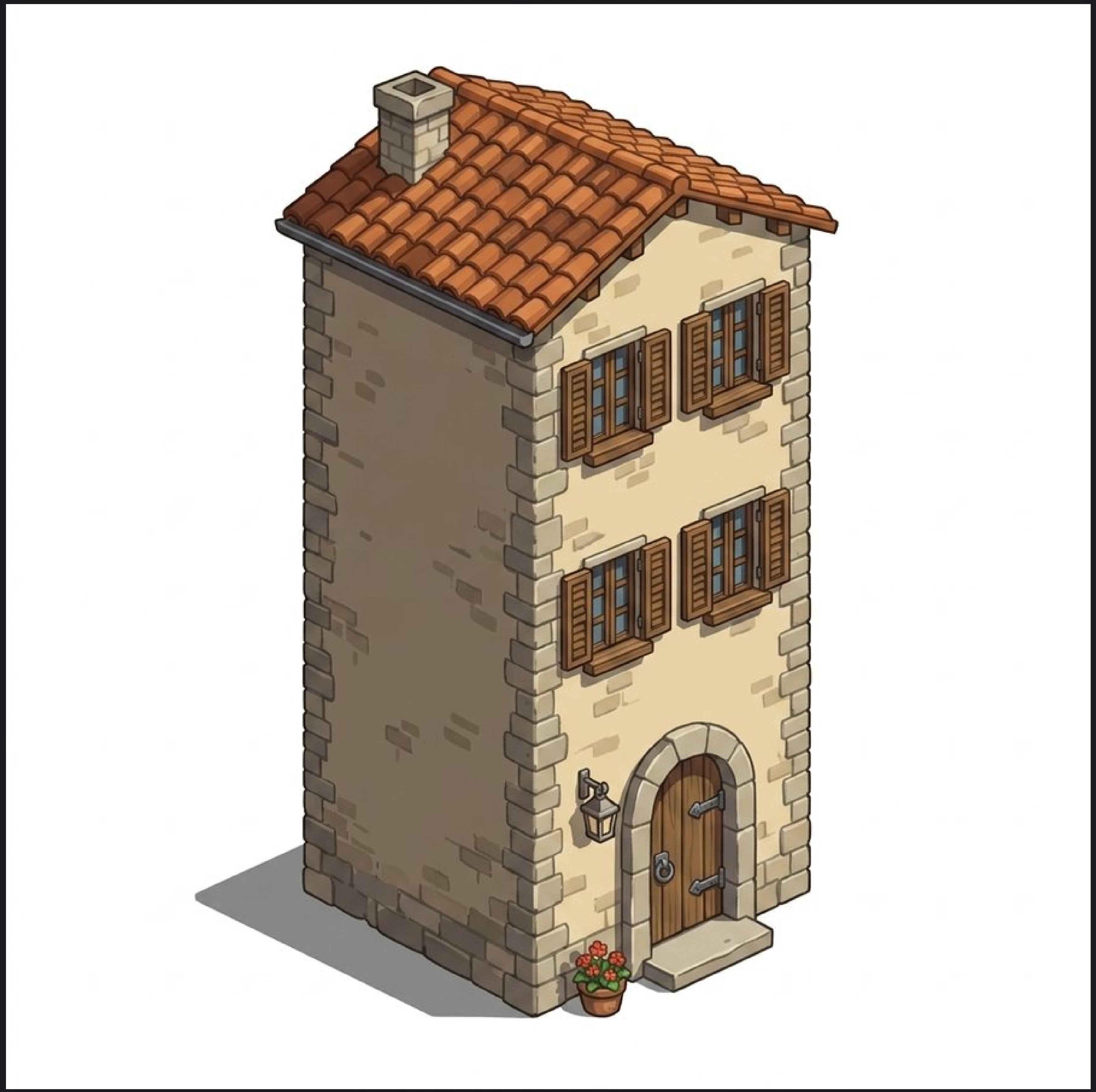
Scaffold Adherence Test: House B (Tower)

Pipeline: Vector Blueprint vs. Diffusion Output | 1024x1024

Deterministic Vector Scaffold (Blueprint)



Generated Result (gemini-3.1-flash-image)



Scaffold Dimensions: Width 3.6m (345px), Wall Height 7.8m (748px), Roof Pitch 1.4m (134px), Chimney on left.

Adherence Assessment: Model successfully generated a 3-storey vertical tower and chimney matching the scaffold layout.

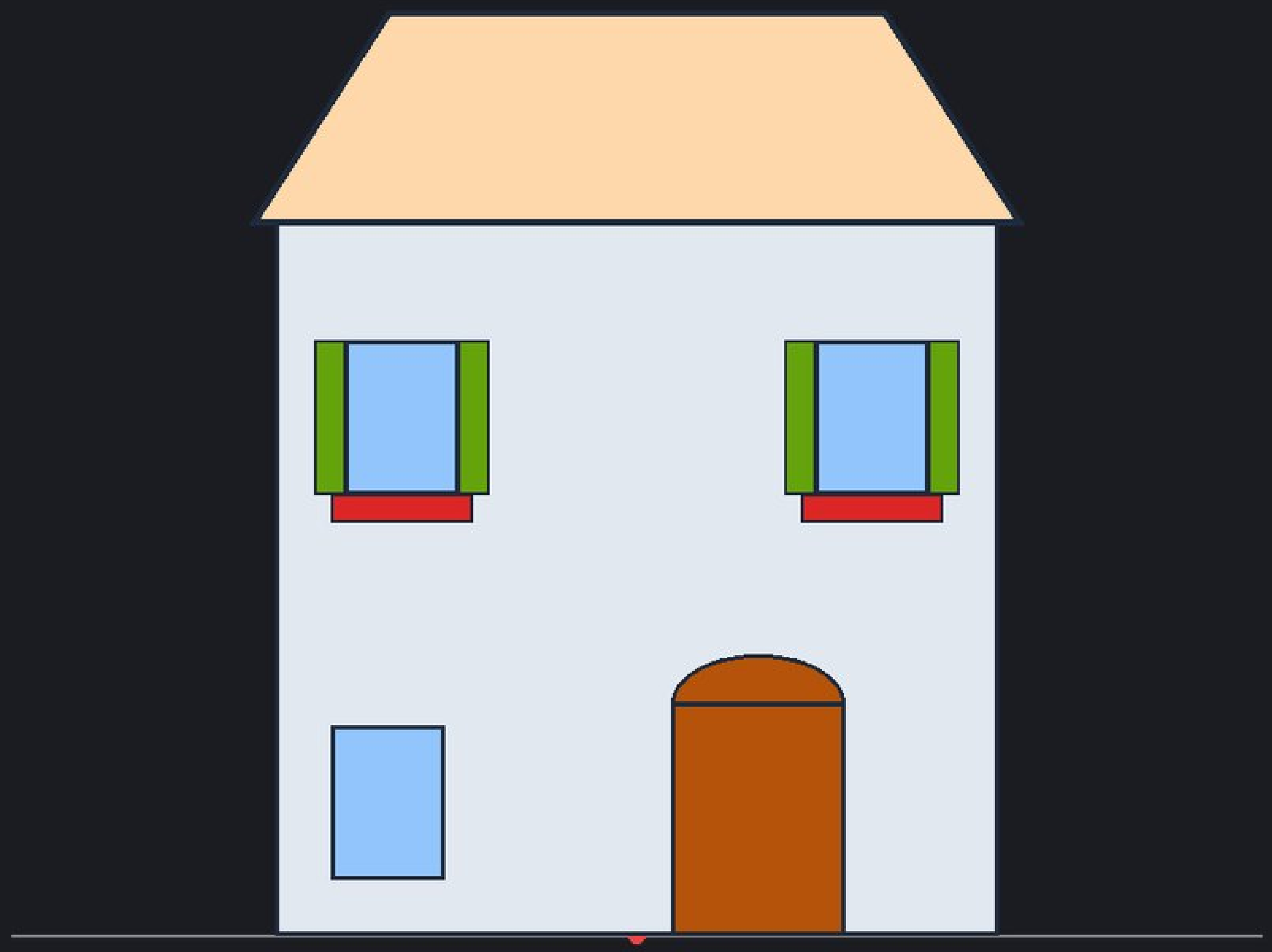
Normalization Factor: x0.985 (Door matched within 1.5% of canonical specification).

SPATIAL CONDITIONING

Scaffold Adherence Test: House C (Shop)

Pipeline: Vector Blueprint vs. Diffusion Output | 1024x1024

Deterministic Vector Scaffold (Blueprint)



Generated Result (gemini-3.1-flash-image)

Scaffold Dimensions: Width 5.4m (518px), Wall Height 5.4m (518px), Hip Roof, Arched Shop Double Door.

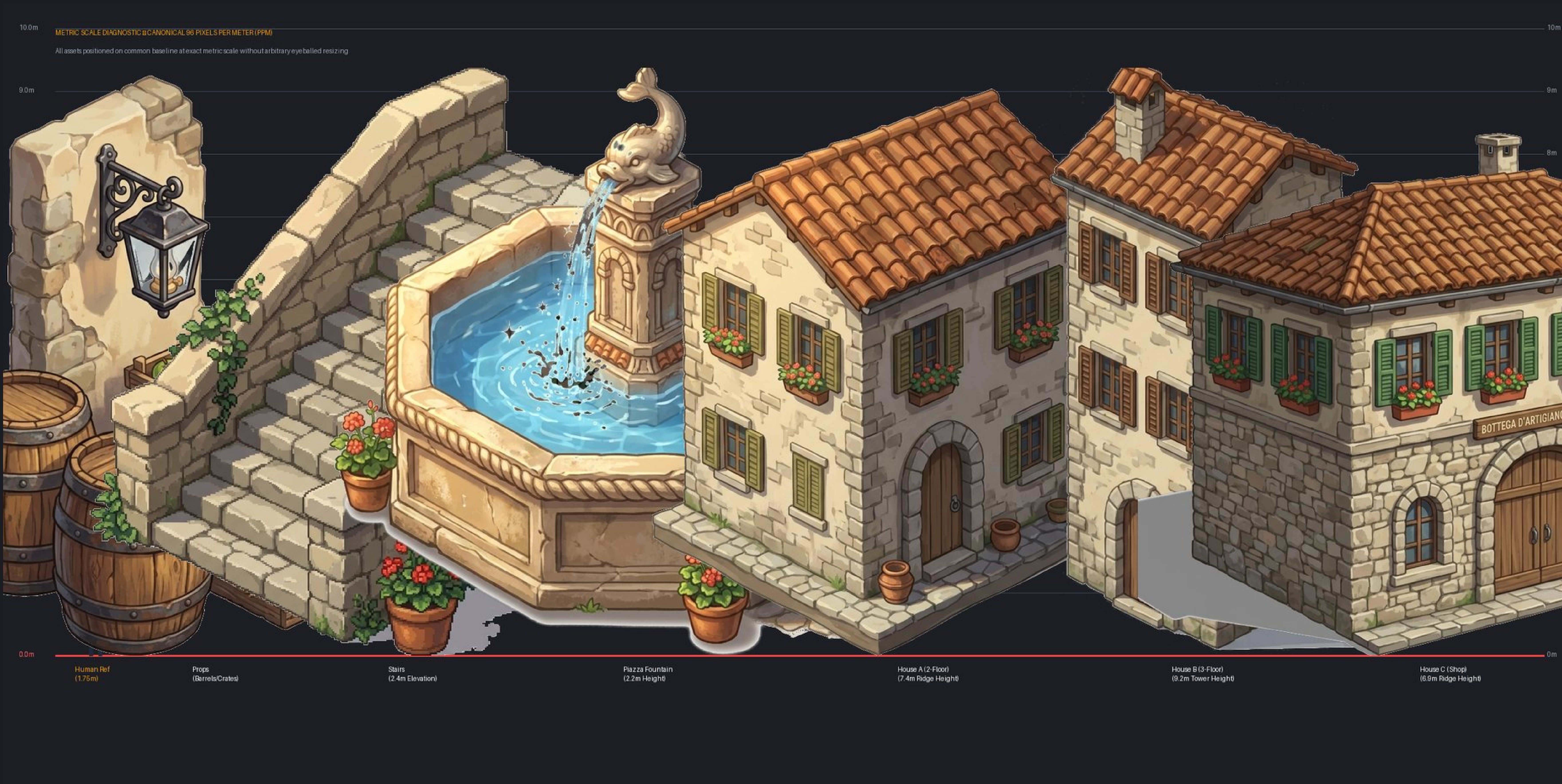
Adherence Assessment: Model produced wide commercial facade with double doors and hip roof corresponding to the blueprint.

Normalization Factor: x0.940 (Door matched within 6% of canonical specification).

METRIC SCALE ANALYSIS

Scale Consistency Diagnostic (Canonical 96 PPM)

Pipeline: Mathematical Metric Verification | Original Resolution: 2400x1200



All normalized assets placed on a common ground baseline alongside a canonical 1.75m (168px) human reference.

Grid Lines: Marked every 1.0 meter (96px). Human ref = 1.75m, Doorway target = 2.10m, Story heights = 2.80m.

Key Observation: Houses A, B, and C adhere to human scale; street props (barrels) are slightly enlarged due to macrogeneration.

DIAGNOSTIC ANALYSIS

Shadow and Perspective Coherence Diagnostics

Pipeline: Analytical Ground & Shadow Analysis | 1920x960

1. Shadow Direction & Angle Variance

2. Ground Axis vs. 2:1 Dimetric Grid (26.565 deg)

Left: Cast shadow azimuth ranges from 120 deg (Fountain) to 145 deg (Props). Highlight: 2D AI lacks unified scene sun coordinates.

Right: Drawn ground slopes range from 24.5 deg to 32.0 deg vs canonical 26.565 deg dimetric slope.

Key Takeaway: Perspective and shadow variance remain the primary obstacle to seamless pure-2D game asset compositing.